Spotify discovering groups

Recursive UMAP and HDBSCAN, explained step by step

InnoIT · ML use case

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Spotify: the problem

Millions of listeners behave differently, but nobody has told us which groups exist

THE PROBLEM

Millions of listeners

Many behaviours each, no predefined groups



Who are they?

HOW IT WORKS

1 · Find groups

UMAP + HDBSCAN, repeated inside a chosen group

group labels

2 · Explain groups

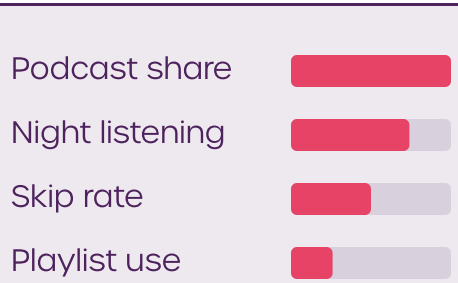
One-vs-all XGBoost + SHAP on the original features

WHAT YOU GET



Groups found

Listeners with similar behaviour. A few stay as noise.



Groups explained

SHAP shows which behaviours define each group, so researchers can validate it.

Clustering proposes the groups. People decide which ones are worth acting on.

Source: Spotify Engineering

Clustering

Spotify (Sweden)

Documented user-research method

Concept

Discover groups with similar behavior without predefined group labels.

Advantage

Finds unexpected segments and supports targeted user research.

Real use case

Spotify combines UMAP with recursive HDBSCAN to discover finer segments of users.

Limitation

Feature choices shape the groups. Stable clusters still need human validation.

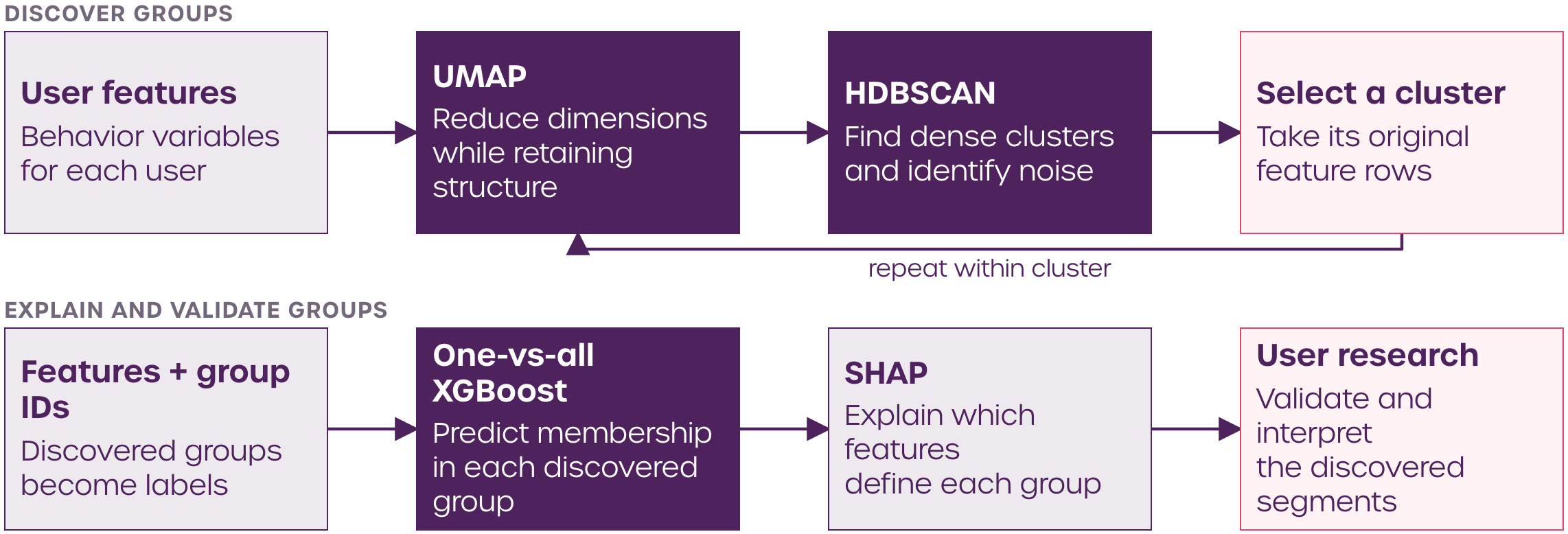
Spotify uses XGBoost and SHAP to explain the resulting segments.

Clustering discovers groups that researchers can then investigate.

Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster



Source: Spotify Engineering

Spotify: discovering and explaining groups

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Spotify

1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 Re-embed | 6 Hierarchy | 7 XGBoost | 8 SHAP | 9 Explainability

The people behind the use case

People	Ana and Luis play AC/DC and Metallica. Marta plays Taylor Swift.
Group	Artist-play counts could group Ana and Luis together, apart from Marta.
Why	An explanation could identify rock and metal listening as the group's main signal.
Possible use	Suggest Iron Maiden to Ana if Luis listens to them.

Illustrative artist example. The computed demo uses behaviour, not artist counts.

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Technical details

A concrete listening example

Ana and Luis share artists. Can we discover their group and explain what makes it different?

Follow the same people

Each next step explains what the algorithm would mean for Ana, Luis and Marta.

Illustrative story. The calculated charts use synthetic behaviour features, without artist counts or named users.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

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Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Original features

$X \in \mathbb{R}^{684 \times 6}$

Unlabelled observations

Listener	Minutes	Sessions	Skips	Discovery	Playlists	Night
L001	222	11.4	10%	17%	52%	72%
L002	124	6.6	46%	64%	41%	29%
L003	32	1.8	28%	27%	75%	8%
L004	127	6.6	42%	66%	32%	36%

Numeric features, no prescribed segment labels

684 rows × 6 features

StandardScaler $z_{ij} = (x_{ij} - \mu_j) / \sigma_j$ Fit once on the full demo population

Computed demo · synthetic listeners

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Technical details

An original feature vector for every listener

Ana's listening becomes a numeric row. Here we compute six behaviour features. Artist counts would be extra inputs.

Listener L001

Minutes	222	Sessions	11.4
Skips	10%	Discovery	17%
Playlists	52%	Night	72%

← / → steps · colours identify local groups · no Spotify data or internal parameters

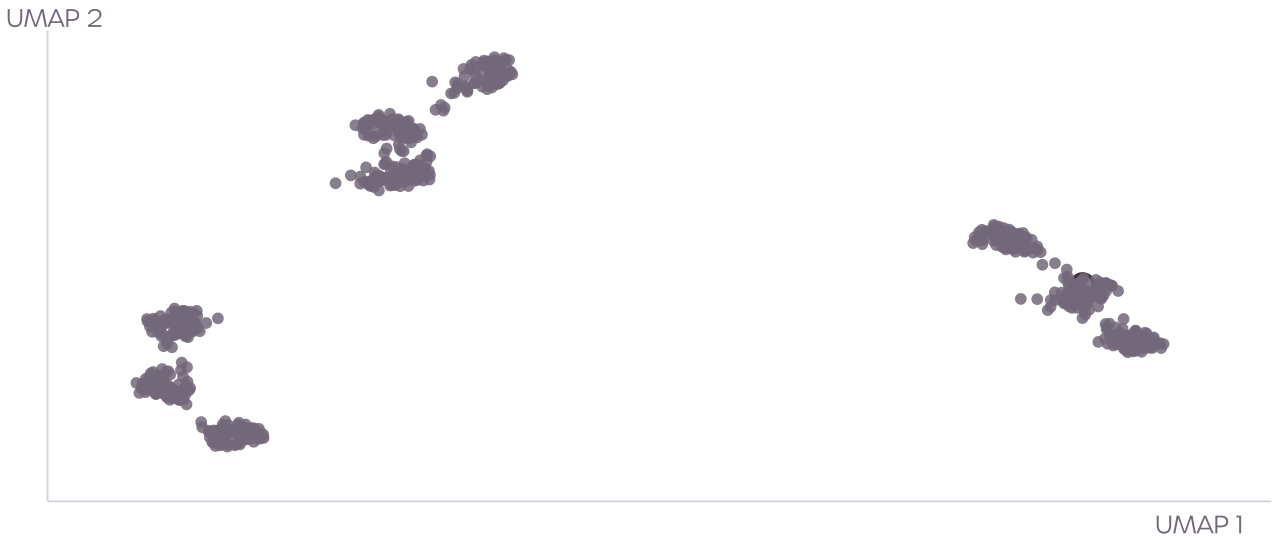
Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Computed UMAP coordinates

3 clusters · 0 noise · 684 listeners



Display transition only. Point movement does not reproduce the optimizer's iterations.

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Technical details

UMAP makes neighbourhoods visible

With suitable artist features, Ana and Luis could appear nearby because they share artists. UMAP displays similarity, not a genre label.

$684 \times 6 \rightarrow 684 \times 2$

10-neighbour trustworthiness: 0.975

Local-neighbourhood diagnostic, not segment validity.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

Spotify: discovering and explaining groups

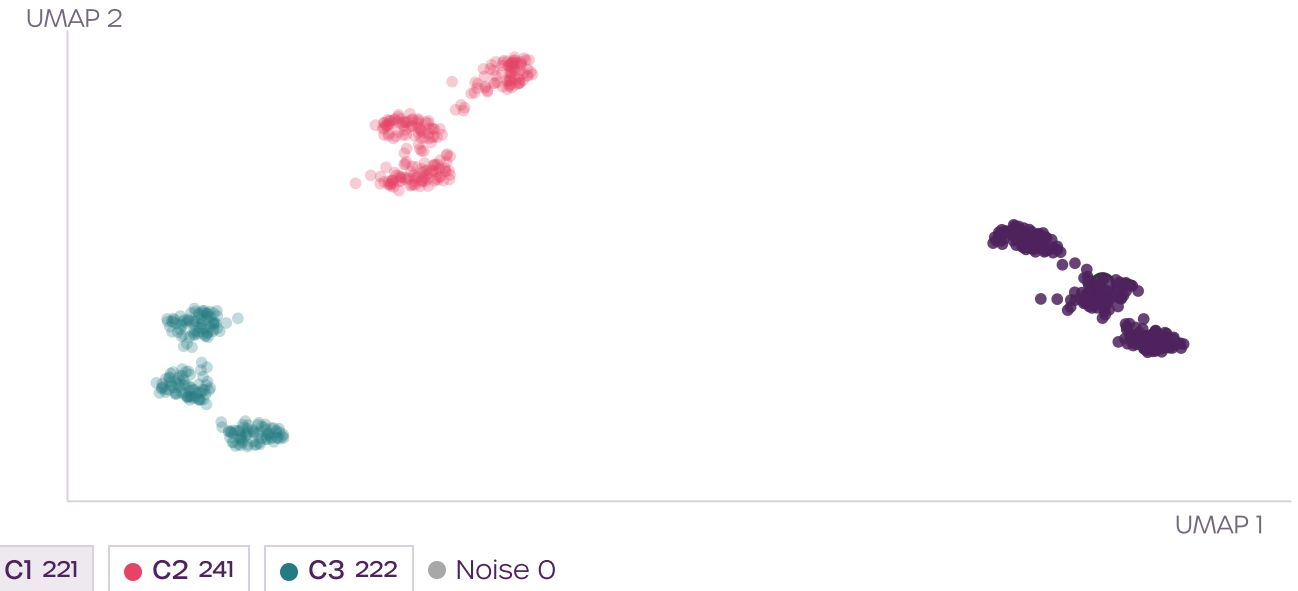
The recursion uses original features within a selected cluster

Spotify

1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 Re-embed | 6 Hierarchy | 7 XGBoost | 8 SHAP | 9 Explainability

Computed UMAP coordinates

3 clusters · 0 noise · 684 listeners



C1 221 | C2 241 | C3 222 | Noise 0

Computed demo · synthetic listeners

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Technical details

HDBSCAN discovers density-based groups

A dense group could contain Ana and Luis, while Marta falls elsewhere. The colours show discovered groups. Our chart uses synthetic behaviour data.

Inspect group: C1 · 221 listeners

32.3% of this node

Noise: 0 / 684

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

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Spotify: discovering and explaining groups

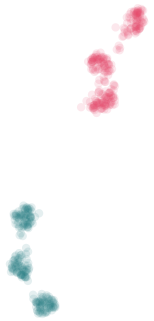
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Spotify

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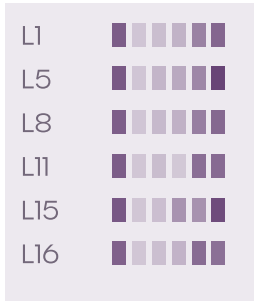
A new fit, not a magnified plot

1. Select C1



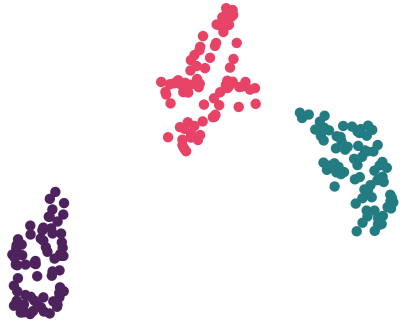


2. Original X rows



221 × 6 features

3. Refit both models



3 groups · 0 noise

New coordinates come from a new UMAP fit on the selected original rows.

Computed demo · synthetic listeners

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Technical details

Recursion starts from the original feature rows

Analyse Ana's group again using its original rows. This could reveal finer habits within her broader group.

Parent cluster: C1 · 221 rows

Inspect C1 child fit

← / → steps · colours identify local groups · no Spotify data or internal parameters

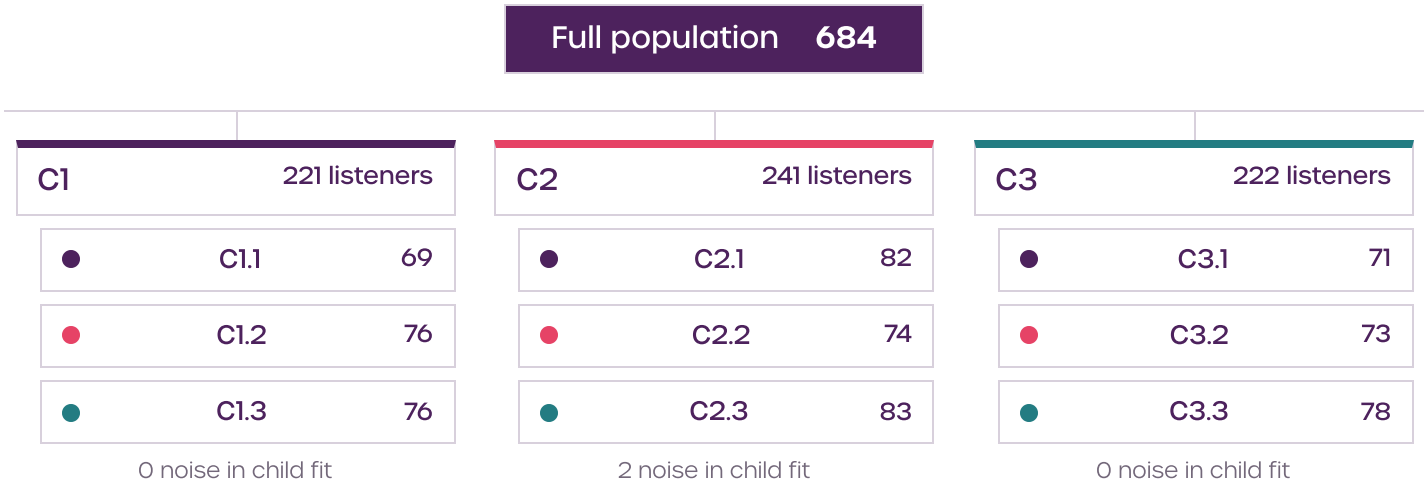
Source: Spotify Engineering

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Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Computed cluster hierarchy



Noise stays unassigned. Child counts can sum to less than their parent.

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Technical details

The hierarchy records each discovery step

Ana's broad group could split into subgroups. The tree records which users enter each new fit. Click a branch to inspect its computed population.

4 computed fits

Each child has its own UMAP coordinates and HDBSCAN assignments.

Computed demo · synthetic listeners

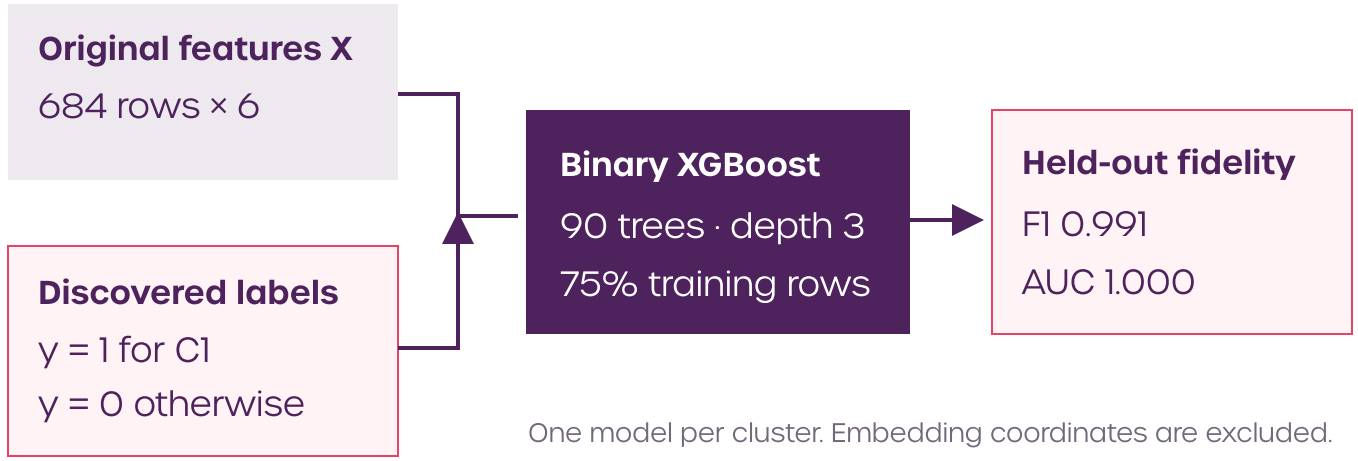
← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

One-versus-rest training



One model per cluster. Embedding coordinates are excluded.

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Technical details

A supervised model learns discovered membership

XGBoost learns to predict Ana's group versus everyone else, using the original features and discovered labels.

Membership model: C1 versus rest

513 training / 171 held-out

The split tests label reproduction, not new-user cluster discovery.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

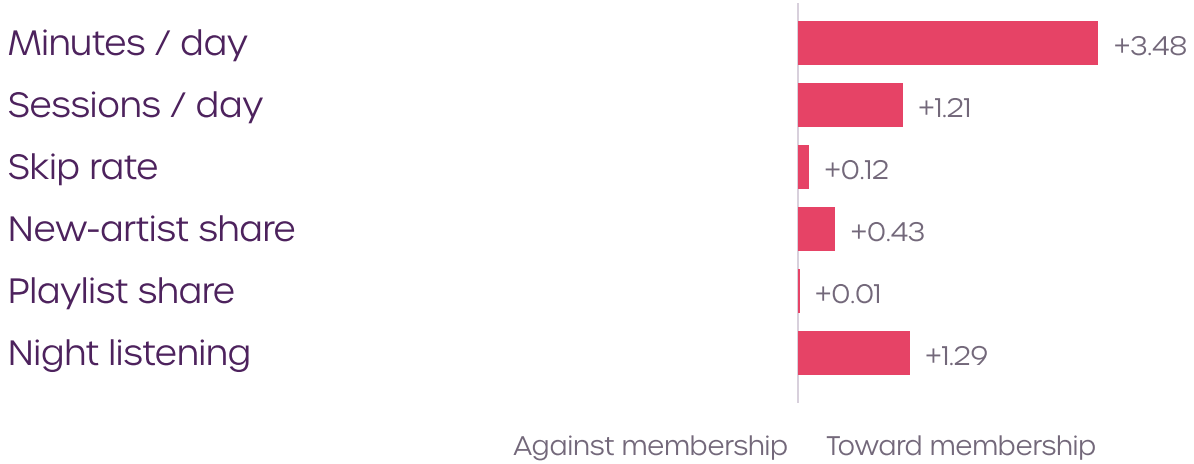
Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Calculated TreeSHAP contributions

This listener | Mean |SHAP| | Signed contribution to membership log-odds



Base -0.80 + Σ SHAP 6.55 = log-odds 5.75

P(C1) = 99.7%

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Technical details

SHAP explains the membership predictor

SHAP explains a listener's group score. Indie listening could matter with artist inputs. These bars explain behaviour.

Explain membership in: C1

Listener: L001 · held-out

Held-out listener. Independent binary scores need not sum to 1.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

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Why interpretation is needed

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Technical details

What the results mean

Group C1 Listener L001

Across listeners

Minutes / day and Night listening have the largest average effects.

This listener

Minutes / day = 222 is the strongest push toward C1.

Model explanations do not prove causes or validate segments.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

Discussion

Which groups could you discover in your own data?

And who would investigate them?

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